

STAT C260: Final Project Report

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1 Introduction and Context

1.1 Motivation

The main topic for investigation in this project concerns the state-level opioid prescribing monitoring laws, specifically Prescription Drug Monitoring Program (PDMP) opioid check mandates in the U.S., which were introduced as a policy response to the opioid epidemic. The opioid epidemic was the rapid increase of opioid misuse beginning in the mid to late 1990s that has led to the deaths of hundreds of thousands of people. These monitoring-based policies had their groundwork earlier, though they reached a turning point around the mid-2010s, after which many states started more rigorously implementing and strengthening PDMP mandates (though this varied state by state).

However, although these policies were well-intentioned, there is speculation that increasing monitoring of prescription opioids may have led some individuals to turn to other illicit drugs as a result. But it is difficult to determine whether these policies directly caused individuals to seek out alternative illicit drugs, since there were also rapid changes in illicit drug supply during this time (including heroin, fentanyl, and other substances). This creates a confounding timeline problem, as it is unclear whether individuals switched to alternatives because of the policy, or whether deaths related to illicit drugs were rising independently due to other factors.

1.2 Causal Question

This motivates the analysis for the following research question: what is the causal effect of implementing a PDMP opioid check mandate on illicit opioid mortality at the state-year level? In particular, the analysis examines whether increased monitoring of prescription opioids through PDMP mandates is associated with changes in illicit opioid mortality, potentially reflecting substitution toward illicit drugs.

2 Conceptual Framework and DAG

Based on this causal question, we can create a DAG as follows:

2.1 DAG

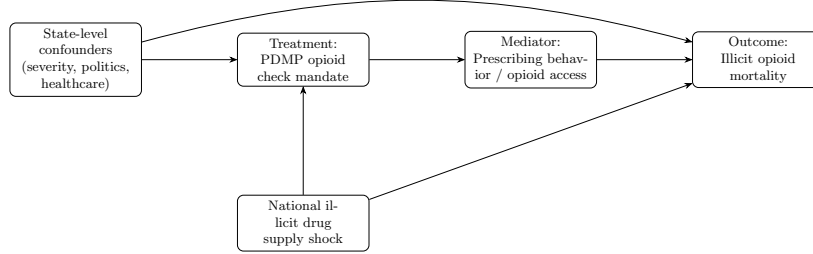


Figure 1: This is a causal DAG for the relationship between PDMP opioid check mandates and illicit opioid mortality.

2.2 DAG Explanation

From the DAG, the PDMP policy is assumed to affect illicit opioid mortality indirectly through changes in prescribing behavior and opioid access. The state-level confounders, such as opioid severity, healthcare conditions, and political factors, may influence both the likelihood of policy adoption and mortality outcomes. Additionally, the national illicit drug supply shocks may affect both treatment assignment and illicit opioid mortality.

These relationships induce back-door paths between treatment and outcome. Therefore, without the additional assumptions, the observed association between PDMP adoption and illicit opioid mortality may reflect both causal effects and confounding factors, potentially complicating identification.

3 Data Explanation

The analysis for this project consists of joining data from two sources: mortality data from the CDC and PDMP policy data for constructing the treatment variable. The first dataset contains opioid mortality data from the CDC, which reports state-level opioid-related death rates over time. In particular, the data distinguishes between prescription opioid mortality and illicit opioid mortality, allowing for separate analysis of these outcomes. Additionally, the second dataset contains information on state-level Prescription Drug Monitoring Program (PDMP) policies, including the timing of implementation of opioid check mandates requiring healthcare professionals to consult the PDMP before issuing prescriptions.

These two datasets are then merged to form a panel of U.S. states observed annually over time. To be more specific: The unit of observation is at the level of state-year, indexed by state i and year t . Then let Y_{it} denote the illicit opioid mortality rate and P_{it} denote the prescription opioid mortality rate in state i and year t . The primary outcome variable is illicit opioid mortality, measured as the crude death rate per state per year.

The key treatment variable is an indicator for whether a PDMP opioid check mandate is in effect in a given state-year. Let G_i denote the first year in which state i implements the policy. The treatment indicator is defined as $D_{it} = \mathbf{1}(t \geq G_i)$, which equals 1 if the policy is active in state i at time t , and 0 otherwise.

3.1 Discussion of the Missing Values

A notable feature of the data is the presence of missing values in the mortality variables, particularly for illicit opioid deaths in earlier years. One thing to observe from inspecting the data is that the missing data is not uniformly distributed across states and time, as earlier years and smaller states exhibit higher rates of suppression in illicit opioid mortality. This suggests that missingness is not random and may be related to underlying outcome levels.

To mitigate this issue, the project focuses on the period after 2010, where data coverage is more comparable across states. These missing observations arise from suppression or unreliability of reported counts in the CDC data. As a result, the analysis restricts attention to the time period after 2010, where data coverage is more consistent across states. This implicitly assumes that remaining missingness does not differentially bias treated and control groups.

The final analysis sample consists of a state-year panel with multiple observations per state. This allows for the use of fixed-effects models to control for time-invariant state characteristics and common time shocks.

4 Empirical Strategy

4.1 Baseline Difference-in-Differences Specification

When first approaching this data, the staggered adoption of PDMP mandates across states may suggest the possibility of treating policy implementation as plausibly exogenous, providing variation suitable for a difference-in-differences design. In particular, differences in policy timing across states could generate identifying variation if adoption were unrelated to underlying outcome trends. To estimate the causal effect of PDMP opioid check mandates on illicit opioid mortality, we can employ a difference-in-differences (DiD) framework using panel data across states and time.

To start, we can let Y_{it} refer to the illicit opioid mortality rate in state i at time t , and also let $D_{it} = \mathbf{1}(t \geq G_i)$ be an indicator equal to 1 if the PDMP mandate is in effect in state i at particular time t , where G_i denotes the year of policy adoption.

To be more specific, the baseline estimating equation is:

$$Y_{it} = \alpha_i + \lambda_t + \beta D_{it} + \varepsilon_{it}$$

where:

- α_i are defined as the state fixed effects, which capture the time-invariant differences across states,

- λ_t are defined as the year fixed effects, which capture the common shocks across all states,
- β is defined as the parameter of interest, representing the average treatment effect of the policy,
- ε_{it} is an error term.

From a causal inference perspective, this regression equation can be interpreted as an implementation of a back-door adjustment, where state and year fixed effects serve as stand-ins for confounding variables that affect both policy adoption and opioid mortality outcomes. To be clear, when considering the DAG, the DiD framework attempts to block back-door paths from treatment to outcome by conditioning on state and time fixed effects, which capture time-invariant state-level confounders and aggregate shocks. However, this approach relies on the assumption that no time-varying confounders remain.

Additionally, one can note that the standard errors are clustered at the state level to account for serial correlation within states over time.

4.2 Identification Assumption

The key identification assumption for the DiD estimator is the parallel trends assumption, which states that in the absence of treatment, the treated and control states would have experienced similar trends in illicit opioid mortality.

To explore this, let $Y_{it}(1)$ and $Y_{it}(0)$ denote the potential outcomes with and without treatment, respectively. So, the causal parameter of interest is:

$$\beta = E[Y_{it}(1) - Y_{it}(0)]$$

Since $Y_{it}(0)$ is not observed for treated states after treatment, the identification relies on the assumption that untreated potential outcomes evolve similarly across treated and untreated states.

This can be expressed as:

$$E[Y_{it}(0) - Y_{is}(0) \mid G_i = g] = E[Y_{jt}(0) - Y_{js}(0) \mid G_j = \infty]$$

for all t, s , where i indexes treated states, and j indexes untreated states.

This assumption seems to imply that that a divergence in outcomes after treatment can be attributed to the policy.

In causal graph terms, this will require that, conditional on state and time effects, there are no unobserved confounders affecting both treatment assignment and potential outcomes (the back-door criterion is satisfied)

4.3 Event Study Specification

To explore the treatment effects and assess the plausibility of the parallel trends assumption, one can utilize an event study specification:

$$Y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -1} \beta_k \mathbf{1}(t - G_i = k) + \varepsilon_{it}$$

where, in this case, k indexes time relative to treatment, and the period $k = -1$ is omitted as the reference category.

The coefficients β_k are now defined to capture the particular effect of the policy at different time periods in relation to adoption. To be more precise, we can define it as the following:

- The pre-treatment coefficients ($k < 0$) will provide a test of the parallel trends assumption,
- The post-treatment coefficients ($k > 0$) will capture the changing effects of the policy.

4.4 Potential Threats to Identification

After analyzing the available data, a potential major concern in this configuration is that policy adoption may actually be endogenous instead of exogenous. States experiencing rising opioid misuse or mortality may be more likely to implement PDMP mandates earlier, violating the parallel trends assumption.

Causally speaking, this can imply that there is a presence of unobserved confounders that simultaneously affect both treatment assignment and outcomes, violating the back-door criterion required for identification.

Additionally, the timing of PDMP adoption overlaps with broader national changes in illicit opioid supply, including the rapid increase in fentanyl availability. These concurrent shocks may differentially affect states and confound the estimated treatment effects.

The presence of non-random missing data in earlier years further complicates identification, as missingness is correlated with state size and outcome levels.

4.5 Robustness and Sensitivity Analyses

To assess the robustness of the results, several additional analyses can be performed as well:

- **Restricted sample analysis:** limiting the sample to early-adopting states to reduce heterogeneity in treatment timing,
- **Heterogeneity analysis:** examining whether treatment effects differ by adoption timing or baseline risk levels,
- **Pre-treatment balance checks:** comparing baseline outcomes between treated and untreated states.

This analysis can help evaluate the credibility of the identification strategy and the robustness of the findings.

5 Descriptive Evidence

We can start by observing the descriptive trends in opioid mortality across states over time. The figure 2 presents the average prescription opioid mortality P_{it} and illicit opioid mortality Y_{it} over the sample period.

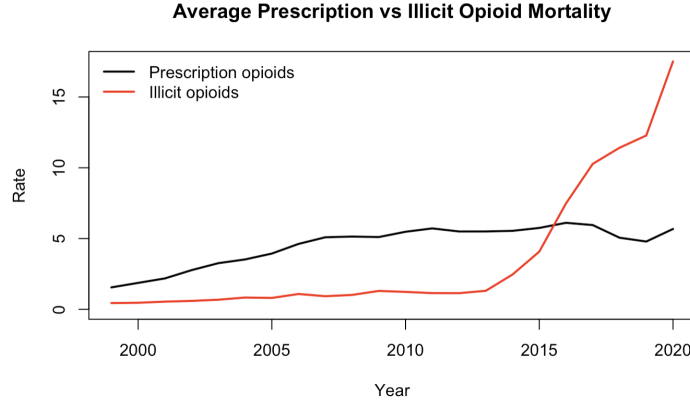


Figure 2: Average prescription and illicit opioid mortality over time. Illicit mortality rises sharply after 2013, while prescription mortality remains relatively stable or declines.

The data reveals that prescription opioid mortality remains relatively stable over time (however with some evidence of decline in later years). In contrast to this, the illicit opioid mortality shows a sharp increase, after around 2013. Interestingly, this period seems to coincide with a major change in the illicit drug supply, including the emergence of synthetic opioids such as fentanyl.

To further examine the differences across the treatment groups, we can also compare the trends in illicit opioid mortality between both the treated and untreated states. To be clear, define D_{it} to the treatment status, and let $D_i = 1$ refer to the states that ever adopt the policy.

It can be observed from the results that the treated states tend to exhibit higher levels of illicit opioid mortality even prior to treatment. Additionally, there also appear to be visible differences in trends between treated and untreated states before policy implementation.

This seems to suggest the following observation

$$E[Y_{it}(0) \mid D_i = 1] \neq E[Y_{it}(0) \mid D_i = 0]$$

before the treatment is applied, perhaps raising concerns about the comparability of treated and control groups.

These patterns indicate potential violations of the parallel trends assumption required for identification.

6 Main Results

Now, the results of the project can be discussed.

6.1 Baseline Difference-in-Differences Estimates

First, an estimate of the baseline difference-in-differences model can be constructed in the following way:

$$Y_{it} = \alpha_i + \lambda_t + \beta D_{it} + \varepsilon_{it}$$

where β can be defined as the following:

$$\beta = E[Y_{it}(1) - Y_{it}(0)]$$

under the particular identifying assumptions mentioned previously.

It can also be seen that the estimated coefficient on D_{it} is positive, however, not statistically significant, when the standard errors are clustered at the same state level.

Table 1: Baseline DiD Estimates of PDMP Effect on Illicit Mortality

	Estimate	Std. Error
Treatment (D_{it})	1.30068	1.57863

This suggests that there is no robust evidence that PDMP mandates have a causal effect on illicit opioid mortality. However, this interpretation is dependent on the identifying assumptions validity.

6.2 Event Study Analysis

To examine the dynamics of treatment effects and test the parallel trends assumption, we can estimate:

$$Y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -1} \beta_k \mathbf{1}(t - G_i = k) + \varepsilon_{it}$$

Figure 3 presents the estimated coefficients β_k .

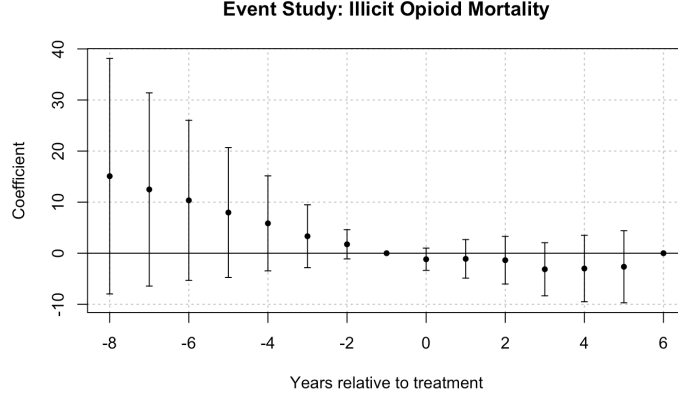


Figure 3: This is the event study estimates of treatment effects. Pre-treatment coefficients deviate from zero, which indicates a violation of the parallel trends assumption.

Several pre-treatment coefficients ($k < 0$) are statistically different from zero, indicating that treated states were already on different trajectories compared to control states prior to policy implementation.

Formally, this implies:

$$E[Y_{it}(0) - Y_{is}(0) \mid D_i = 1] \neq E[Y_{jt}(0) - Y_{js}(0) \mid D_j = 0]$$

which violates the parallel trends assumption.

It can be noted here that the post-treatment coefficients do not display a consistent or stable pattern, further complicating interpretation.

6.3 State-Specific Trend Model

As an additional robustness check, we can also estimate the following:

$$Y_{it} = \alpha_i + \lambda_t + \delta_i t + \beta D_{it} + \varepsilon_{it}$$

where $\delta_i t$ captures the particular state-specific linear trends.

Table 2: DiD Estimates with State-Specific Trends

	Estimate	Std. Error
Treatment (D_{it})	-0.494453	0.382731

Here, the estimated effect becomes smaller and changes sign, suggesting that baseline results may reflect differential trends rather than a causal effect.

6.4 Sensitivity to Sample Restrictions

We can restrict the sample to early-adopting states to attempt to reduce the observed variation in treatment timing.

Table 3: DiD Estimates: Early Adopters Only

	Estimate	Std. Error
Treatment (D_{it})	4.63105	2.65881

Here, the estimated effect is larger but imprecise, indicating sensitivity to sample selection and further weakening causal interpretation.

The estimated effects seem to vary across specifications, indicating sensitivity to sample selection.

6.5 Summary

Taken together, this analysis suggests that $\hat{B}$ is sensitive to the specification and sample choices.

Combined with the previously mentioned evidence of pre-treatment trends, this raises concerns about the credibility of causal interpretation in this case.

7 Diagnosis

The analysis of the empirical results presented above reveal that the identifying assumptions required for causal interpretation of the DiD estimator seem unlikely to hold in this particular configuration.

To be more specific, the event study analysis reveals the presence of non-zero pre-treatment coefficients, indicating that treated and control states exhibit differential trends in illicit opioid mortality even prior to policy adoption. This appears to violate the parallel trends assumption and suggests that untreated potential outcomes are not evolving similarly across groups.

Additionally, the descriptive evidence seems to reveal that treated states tend to have higher levels of illicit opioid mortality relative to untreated states, even before treatment occurs. This may indicate that the presence of selection into treatment, where states experiencing more severe opioid conditions, are more likely to adopt PDMP mandates.

Finally, we can note that the timing of policy adoption is highly clustered in the mid-2010s, which coincides with a period of rapidly increasing illicit opioid deaths driven in part by the emergence of fentanyl. This interesting overlap suggests that policy implementation may be a response to increasingly worse outcomes rather than an outside, exogenous intervention.

Utilizing these observations, these findings imply that the treatment assignment is endogenous. To be more clear, there seem to be unobserved confounders that affect both the policy adoption and illicit opioid mortality, which violates

the back-door criterion required for identification. As a result, the estimated treatment effects cannot necessarily be interpreted as causal effects.

Rather than providing evidence of a true zero effect, the results instead highlight the difficulty of isolating causal effects in a setting where policy adoption is closely tied to underlying outcome trends.

8 Extension and Alternative Design

As mentioned previously, there are identification challenges associated with PDMP policy adoption, so it may be more beneficial to consider an alternative source of variation that is plausibly exogenous.

One potential idea can be to utilize the timing of the rapid increase in fentanyl availability beginning around 2013 as a natural experiment. This shock represents a major change in the illicit opioid market that is not directly driven by the state-level policy decisions. An analysis based on this variation could better isolate the effects of changes in drug supply on illicit opioid mortality.

Another approach can be to examine substitution patterns between prescription and illicit opioids. By analyzing the joint trends in the prescription opioid mortality and illicit opioid mortality, it may be possible to assess whether reductions in prescription-related harm are accompanied by increases in illicit drug use. Although this may not fully account for the identification concerns, it can be useful for additional insight into analyzing the underlying observed trends.

9 Conclusion

The purpose of this project was intended to investigate the causal effect of PDMP opioid check mandates on illicit opioid mortality using a difference-in-differences framework applied to state-year panel data.

The results showed that the empirical results do not provide credible evidence of a causal effect of PDMP rulings on illicit opioid mortality. While the baseline estimates (image referenced previously) suggest a positive association, these results were not robust to clustering and are sensitive to model specification. Moreover, the event study analysis reveals clear violations of the parallel trends assumption, and additional evidence indicates the presence of endogenous policy adoption.

These findings suggest that states perhaps implemented the PDMP mandates in response to worsening opioid conditions, rather than independently of the underlying trends. As a result, the identifying assumptions required for causal interpretation are not likely to hold, and the estimated effects should be interpreted with caution.

At a high level, this analysis seems to reveal the potential challenges of causal policy evaluation in observational settings where treatment is not randomly assigned. In the context of the opioid epidemic, the overlapping policy

interventions and rapid changes in illicit drug supply make it difficult to separate the causal relationships.

In the future, we can focus on alternative research designs that exploit more exogenous sources of variation, as well as improved data that better captures underlying mechanisms. Despite these limitations, the analysis provides some insight into the complexities of evaluating policy interventions in the evolving opioid crisis.

9.1 Code

The code can be found here:

https://drive.google.com/drive/folders/1Rh9QT_YlXKgaycVhD77DqYa0rbZcKpY1?usp=drive_link